

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler

df = pd.read_csv('heart_failure_clinical_records_dataset.csv')
display(df.head())
display(df.describe())
```

	age	anaemia	creatinine_phosphokinase	diabetes
0	75.0	0	582.0	0
20				
1	55.0	0	7861.0	0
38				
2	65.0	0	146.0	0
20				
3	50.0	1	111.0	0
20				
4	65.0	1	160.0	1
20				

	high_blood_pressure	platelets	serum_creatinine	serum_sodium	sex
0	1	265000.00	1.9	130	1
1	0	263358.03	1.1	136	1
2	0	162000.00	1.3	129	1
3	0	210000.00	1.9	137	1
4	0	327000.00	2.7	116	0

	smoking	time	DEATH_EVENT
0	0	4	1.0
1	0	6	1.0
2	1	7	1.0
3	0	7	1.0
4	0	8	1.0

	age	anaemia	creatinine_phosphokinase	diabetes
count	298.000000	299.000000	298.000000	299.000000
mean	60.803134	0.431438	583.486577	0.418060
std	11.902900	0.496107	971.501186	0.494067
min	40.000000	0.000000	23.000000	0.000000
25%	51.000000	0.000000	118.250000	0.000000
50%	60.000000	0.000000	250.000000	0.000000
75%	70.000000	1.000000	582.000000	1.000000

max	95.000000	1.000000	7861.000000	1.000000
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	ejection_fraction	high_blood_pressure	platelets	\
count	299.000000	299.000000	298.000000	
mean	38.083612	0.351171	263358.029262	
std	11.834841	0.478136	97968.752102	
min	14.000000	0.000000	25100.000000	
25%	30.000000	0.000000	212250.000000	
50%	38.000000	0.000000	262000.000000	
75%	45.000000	1.000000	303750.000000	
max	80.000000	1.000000	850000.000000	

	serum_creatinine	serum_sodium	sex	smoking
time \				
count	299.000000	299.000000	299.000000	299.000000
mean	1.39388	136.625418	0.648829	0.32107
std	1.03451	4.412477	0.478136	0.46767
min	0.50000	113.000000	0.000000	0.00000
25%	0.90000	134.000000	0.000000	0.00000
50%	1.10000	137.000000	1.000000	0.00000
75%	1.40000	140.000000	1.000000	1.00000
max	9.40000	148.000000	1.000000	1.00000

	DEATH_EVENT
count	298.000000
mean	0.318792
std	0.466792
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000
max	1.000000

```
print("Shape: ", df.shape)
display("rows:", df[0:0])
display(df.dtypes.to_frame('Data Types'))
display(df.isna().sum().to_frame('missing'))
```

Shape: (299, 13)

'rows:'

```
Empty DataFrame
```

```
Columns: [age, anaemia, creatinine_phosphokinase, diabetes,  
ejection_fraction, high_blood_pressure, platelets, serum_creatinine,  
serum_sodium, sex, smoking, time, DEATH_EVENT]
```

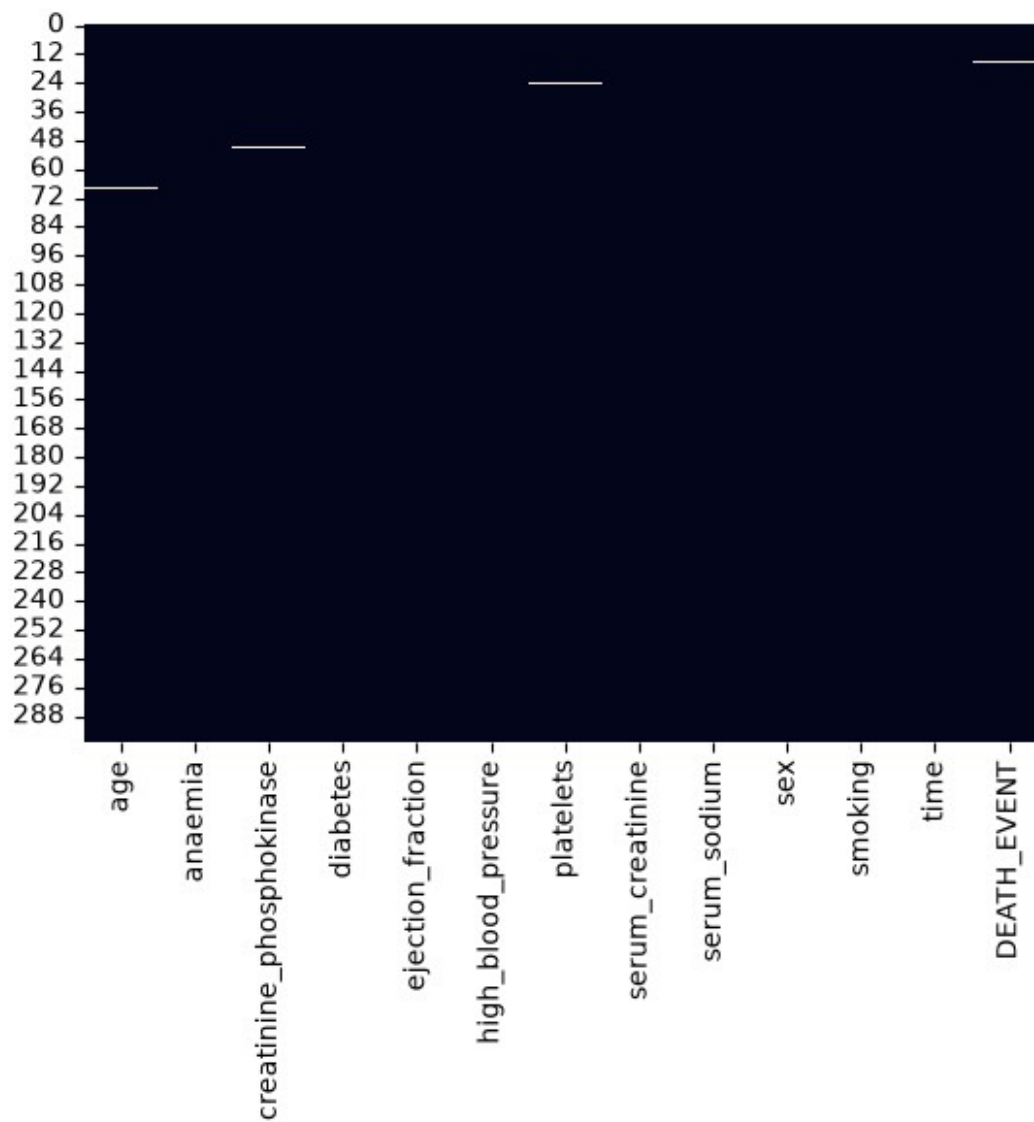
```
Index: []
```

	Data Types
age	float64
anaemia	int64
creatinine_phosphokinase	float64
diabetes	int64
ejection_fraction	int64
high_blood_pressure	int64
platelets	float64
serum_creatinine	float64
serum_sodium	int64
sex	int64
smoking	int64
time	int64
DEATH_EVENT	float64

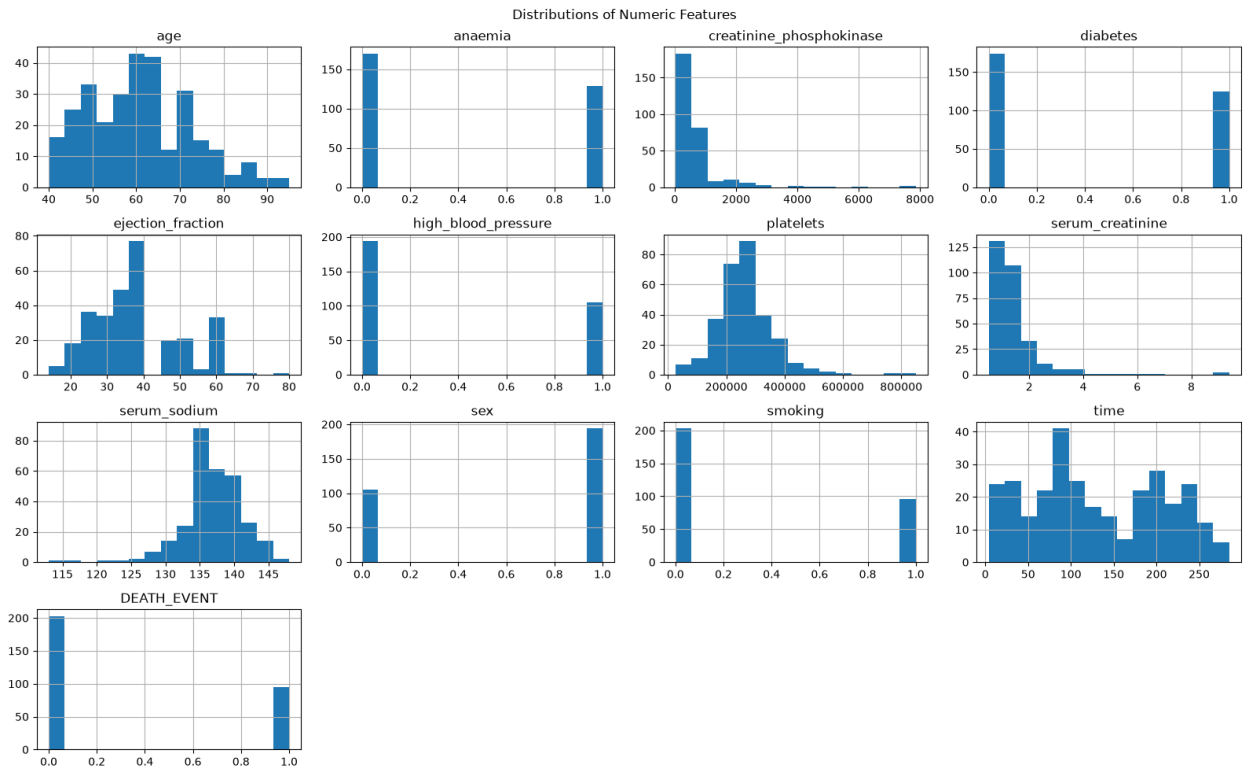
	missing
age	1
anaemia	0
creatinine_phosphokinase	1
diabetes	0
ejection_fraction	0
high_blood_pressure	0
platelets	1
serum_creatinine	0
serum_sodium	0
sex	0
smoking	0
time	0
DEATH_EVENT	1

```
sns.heatmap(df.isnull(), cbar=False)
```

```
plt.show()
```

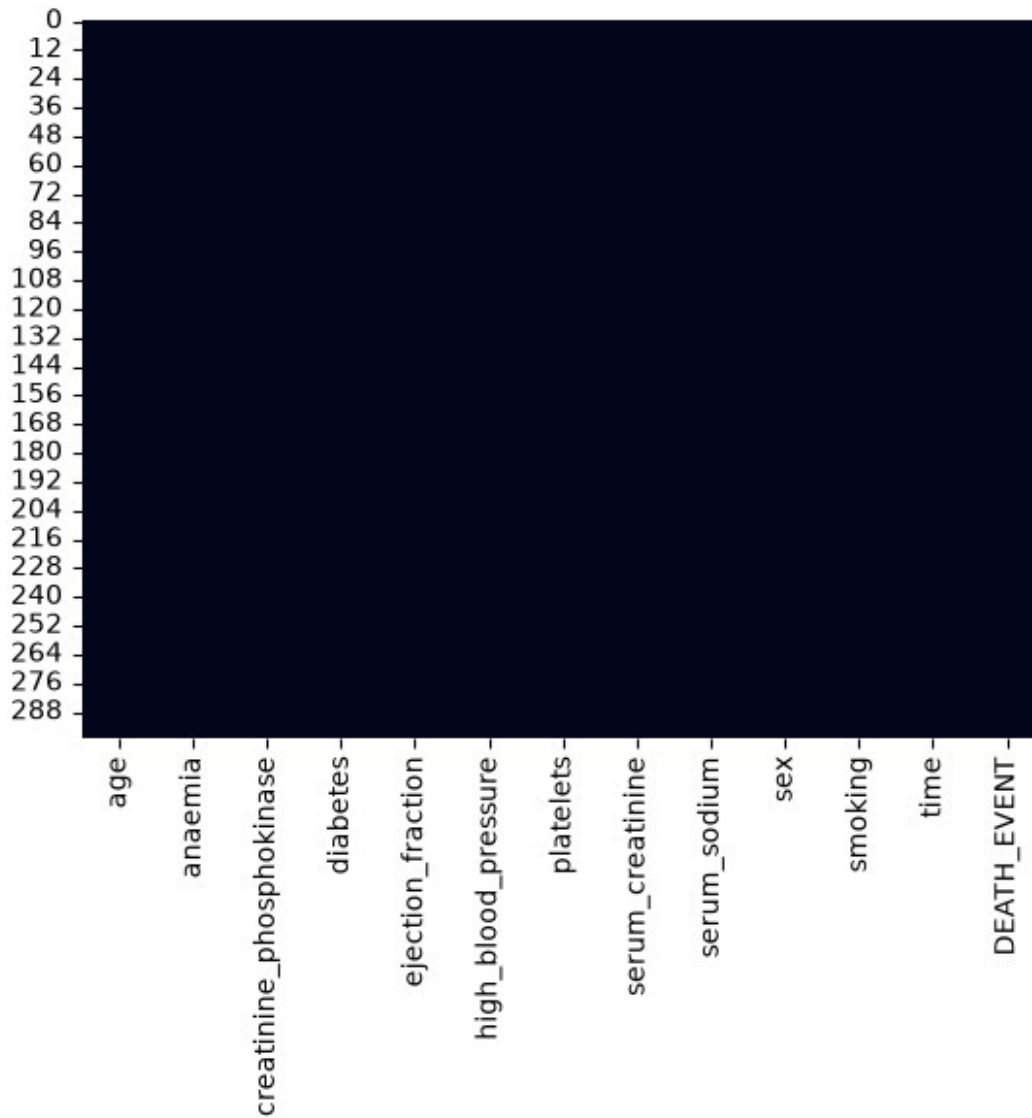


```
df.hist(figsize=(16, 10), bins=15)
plt.suptitle("Distributions of Numeric Features")
plt.tight_layout()
plt.show()
```



```
df_clean = df.fillna(df.mean())
sns.heatmap(df_clean.isna(), cbar=False)
len(df_clean)
```

299



```
Q1 = df_clean.quantile(0.25)
Q3 = df_clean.quantile(0.75)
IQR = Q3 - Q1
print(len(df_clean))
df_clean = df_clean[~((df_clean < (Q1 - 1.5 * IQR)) | (df_clean > (Q3 + 1.5 * IQR))).any(axis=1)]
print(len(df_clean))

299
224

print(df[df['age'] < 0])
```

Empty DataFrame

Columns: [age, anaemia, creatinine_phosphokinase, diabetes, ejection_fraction, high_blood_pressure, platelets, serum_creatinine,

```
serum_sodium, sex, smoking, time, DEATH_EVENT]
Index: []
```

```
df_clean['age_group'] = pd.cut(
    df_clean['age'],
    bins=[0, 40, 60, float('inf')],
    labels=[0, 1, 2],
    right=False
)
df_clean['risk_score'] = df_clean['ejection_fraction'] *
df_clean['serum_creatinine']
df_clean['risk_score'] = (df_clean['risk_score'] -
df_clean['risk_score'].min()) / (df_clean['risk_score'].max() -
df_clean['risk_score'].min())
df_clean.head()
print(df_clean.isna().sum())
```

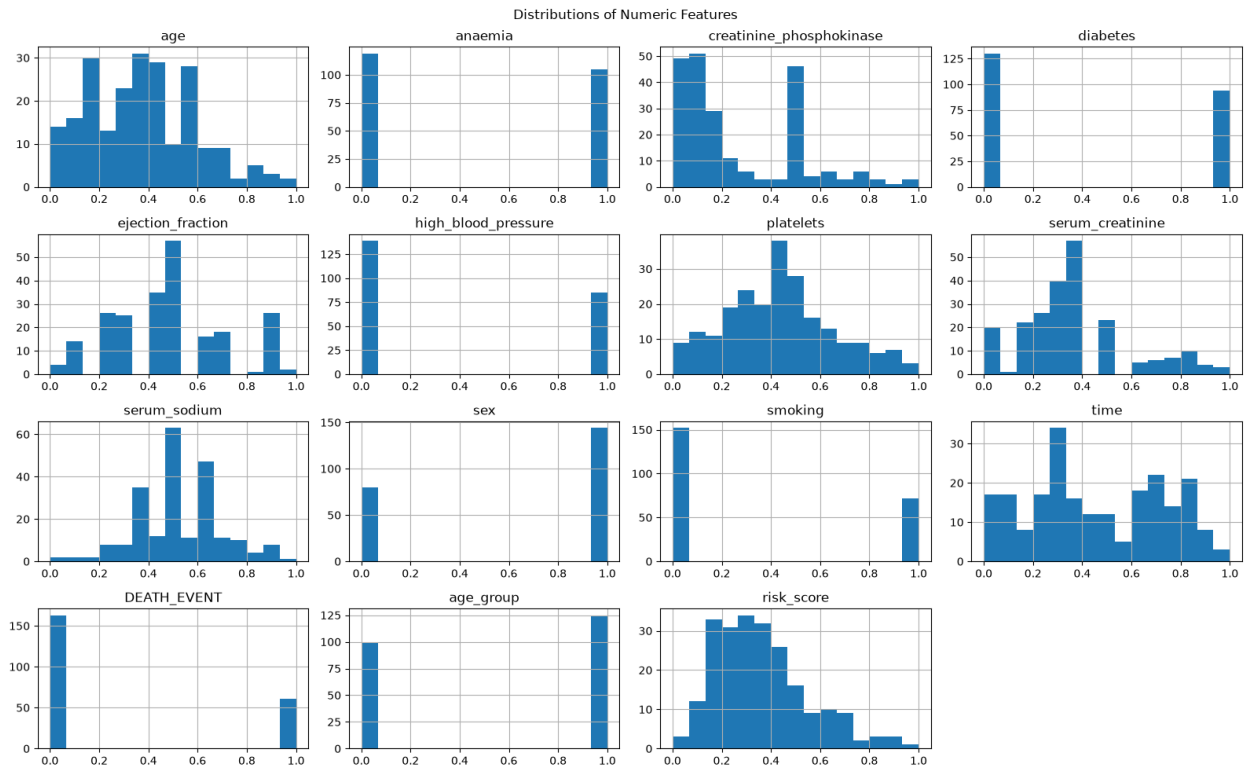
```
age                0
anaemia            0
creatinine_phosphokinase  0
diabetes           0
ejection_fraction  0
high_blood_pressure  0
platelets          0
serum_creatinine   0
serum_sodium       0
sex                0
smoking            0
time               0
DEATH_EVENT        0
age_group          0
risk_score         0
dtype: int64
```

```
scaler = MinMaxScaler().set_output(transform="pandas")
df_clean = pd.DataFrame(
    scaler.fit_transform(df_clean),
    columns=df_clean.columns,
    index=df_clean.index
)
df_clean.describe()
```

	age	anaemia	creatinine_phosphokinase	diabetes	\
count	224.000000	224.000000	224.000000	224.000000	
mean	0.377609	0.46875	0.263772	0.419643	
std	0.216912	0.50014	0.238494	0.494606	
min	0.000000	0.00000	0.000000	0.000000	
25%	0.181818	0.00000	0.072526	0.000000	
50%	0.363636	0.00000	0.155717	0.000000	
75%	0.545455	1.00000	0.470990	1.000000	

max	1.000000	1.00000		1.000000	1.000000
ejection_fraction high_blood_pressure platelets					
serum_creatinine \					
count	224.000000		224.000000	224.000000	
224.000000					
mean	0.474090		0.379464	0.438902	
0.352381					
std	0.229624		0.486340	0.220176	
0.218530					
min	0.000000		0.000000	0.000000	
0.000000					
25%	0.313725		0.000000	0.288525	
0.200000					
50%	0.470588		0.000000	0.460656	
0.333333					
75%	0.607843		1.000000	0.574590	
0.466667					
max	1.000000		1.000000	1.000000	
1.000000					
serum_sodium sex smoking time					
DEATH_EVENT \					
count	224.000000	224.000000	224.000000	224.000000	224.000000
mean	0.523680	0.642857	0.321429	0.457836	0.272321
std	0.165510	0.480231	0.468071	0.272526	0.446151
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.391304	0.000000	0.000000	0.255338	0.000000
50%	0.521739	1.000000	0.000000	0.412811	0.000000
75%	0.652174	1.000000	1.000000	0.715302	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000
age_group risk_score					
count	224.000000	224.000000			
mean	0.558036	0.360744			
std	0.497733	0.187138			
min	0.000000	0.000000			
25%	0.000000	0.217845			
50%	1.000000	0.319235			
75%	1.000000	0.449594			
max	1.000000	1.000000			

```
df_clean.hist(figsize=(16, 10), bins=15)
plt.suptitle("Distributions of Numeric Features")
plt.tight_layout()
plt.show()
```



```
display(df_clean.isna().sum().to_frame('missing'))
print(df_clean.shape)
```

	missing
age	0
anaemia	0
creatinine_phosphokinase	0
diabetes	0
ejection_fraction	0
high_blood_pressure	0
platelets	0
serum_creatinine	0
serum_sodium	0
sex	0
smoking	0
time	0
DEATH_EVENT	0
age_group	0
risk_score	0

(224, 15)

